Western Governors University: MSDA Program

# Scalable Data Engineering Pipeline for Fraud Detection in Financial Transactions

Course: Data Engineering Capstone

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# Scalable Data Engineering Pipeline for Fraud Detection in Financial Transactions

## A. Research Question

The purpose of this project is to address the following research question: *How can financial institutions detect fraudulent transactions more accurately and efficiently using a scalable data engineering pipeline and machine learning?*

Financial institutions process millions of transactions daily, making fraud detection a critical challenge (Ngai et al., 2011). Traditional methods often struggle to keep up with the volume and complexity of transactional data, resulting in delayed detection and increased financial losses. A scalable data engineering pipeline can improve data processing efficiency, while machine learning techniques can enhance the ability to identify fraudulent behavior.

The hypothesis for this study is that a scalable data engineering pipeline combined with machine learning will improve fraud detection to a higher accuracy compared to using a machine learning model alone.

## B. Data Collection

The dataset used in this project was obtained from a publicly available Kaggle dataset (Manchanda, n.d.). It contains anonymized financial transaction records, including transaction type, transaction amount, account balances, and indicators of fraudulent activity.

An advantage of this dataset is its large size and simulation of real-world financial transactions, making it suitable for fraud detection analysis. A disadvantage is that the dataset is simulated and may not fully capture the complexity of real-world fraud patterns.

One challenge encountered during data collection was the dataset's highly imbalanced nature, with fraudulent transactions accounting for only a small percentage of the total data. This challenge was addressed by focusing on evaluation metrics such as recall and precision, which are more appropriate for imbalanced classification problems.

Figure 1. Distribution of fraudulent and non-fraudulent transactions, highlighting the significant class imbalance present in the dataset.

As shown in Figure 1, fraudulent transactions represent a very small proportion of the dataset, which impacts model performance and necessitates the use of evaluation metrics such as recall.

**C. Data Extraction and Preparation**

Data extraction and preparation were performed using Python within the IntelliJ integrated development environment. The Pandas library was used to load and manipulate the dataset.

The dataset was first loaded from a CSV file and inspected to understand its structure. Missing values were removed, and duplicate records were eliminated to ensure data quality. Categorical variables, such as transaction type, were transformed using one-hot encoding to make them suitable for machine learning analysis. Additionally, non-informative identifier columns were removed to improve model performance.

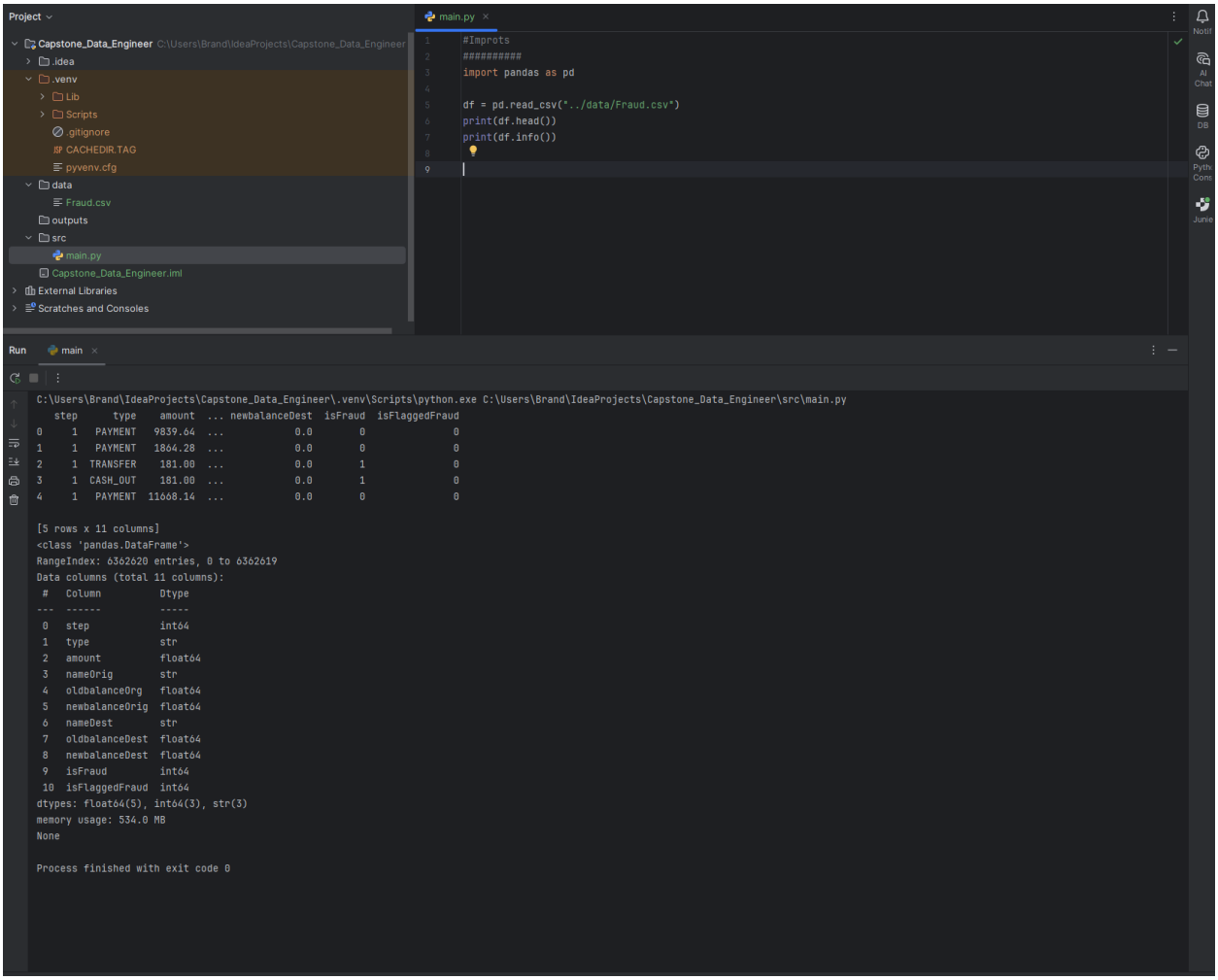


Figure 2. Dataset preview and structure after loading into Pandas.

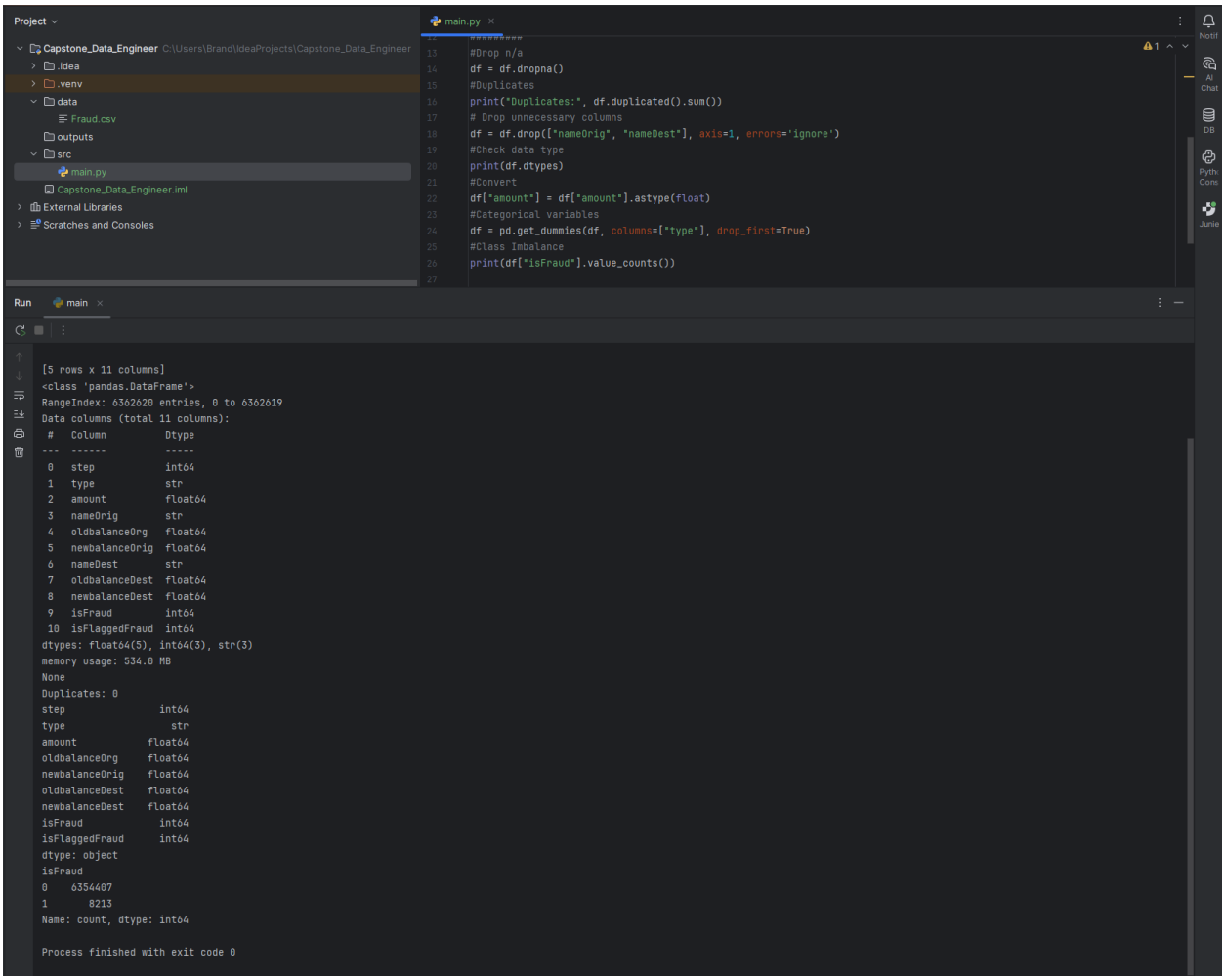


Figure 3. Data cleaning, duplicate removal, data type verification, and dataset class distribution.

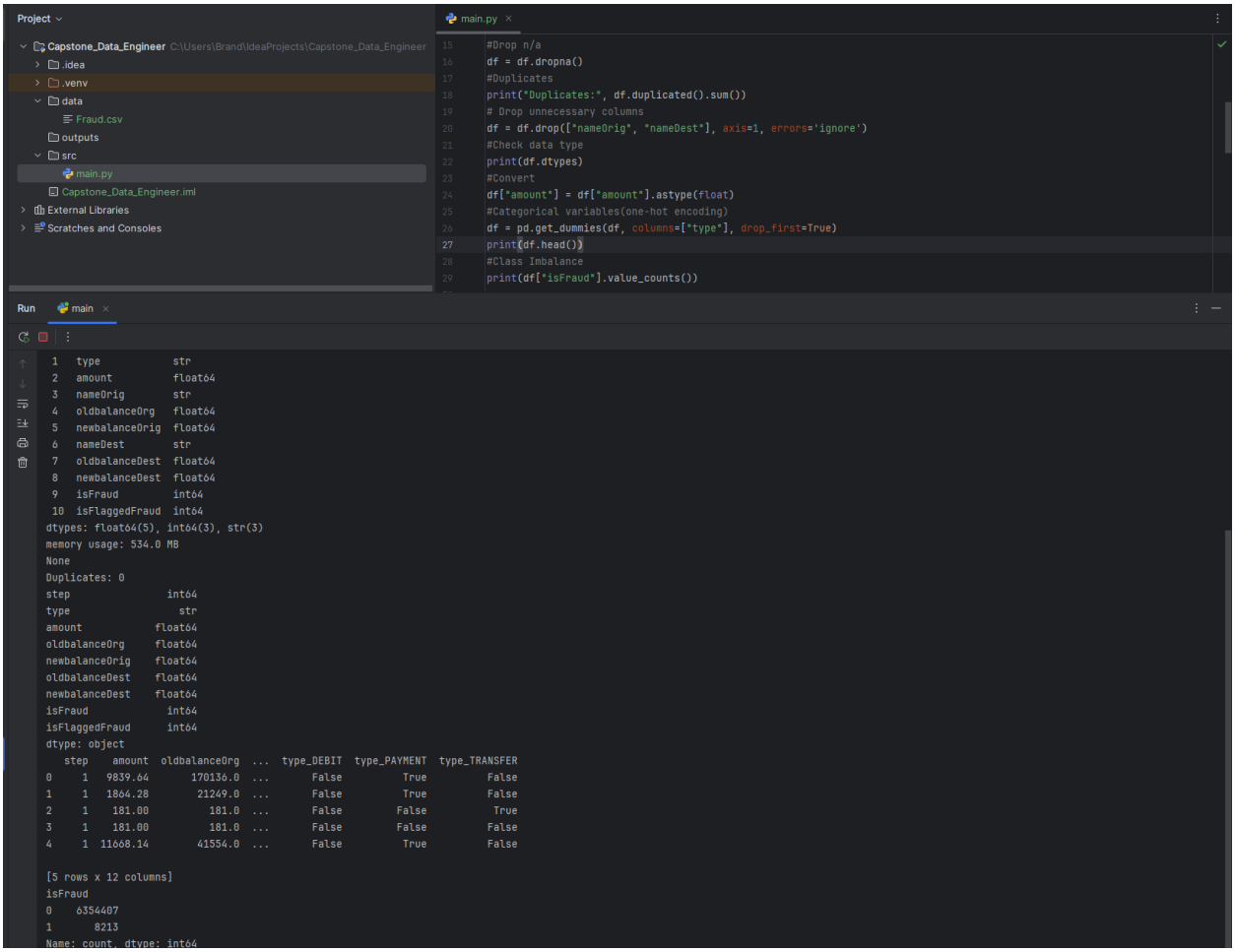


Figure 4. Dataset after one-hot encoding of categorical variables for machine learning analysis.

The categorical variable “type” was converted into multiple binary columns using one-hot encoding to allow the machine learning model to process the data effectively.

These tools and techniques were selected because they offer efficient, scalable methods for handling structured data. An advantage of using Pandas is its ability to process large datasets efficiently, while a disadvantage is that it may encounter memory limitations when working with very large datasets.

## D. Analysis

Logistic regression was used as the primary analysis technique to classify transactions as fraudulent or non-fraudulent. This method was selected because fraud detection is a binary classification problem, and logistic regression is a simple, interpretable model well-suited to such tasks.

The dataset was split into training and test sets, and the model was trained on the processed data. Predictions were generated and evaluated using performance metrics including accuracy, precision, recall, and a confusion matrix.

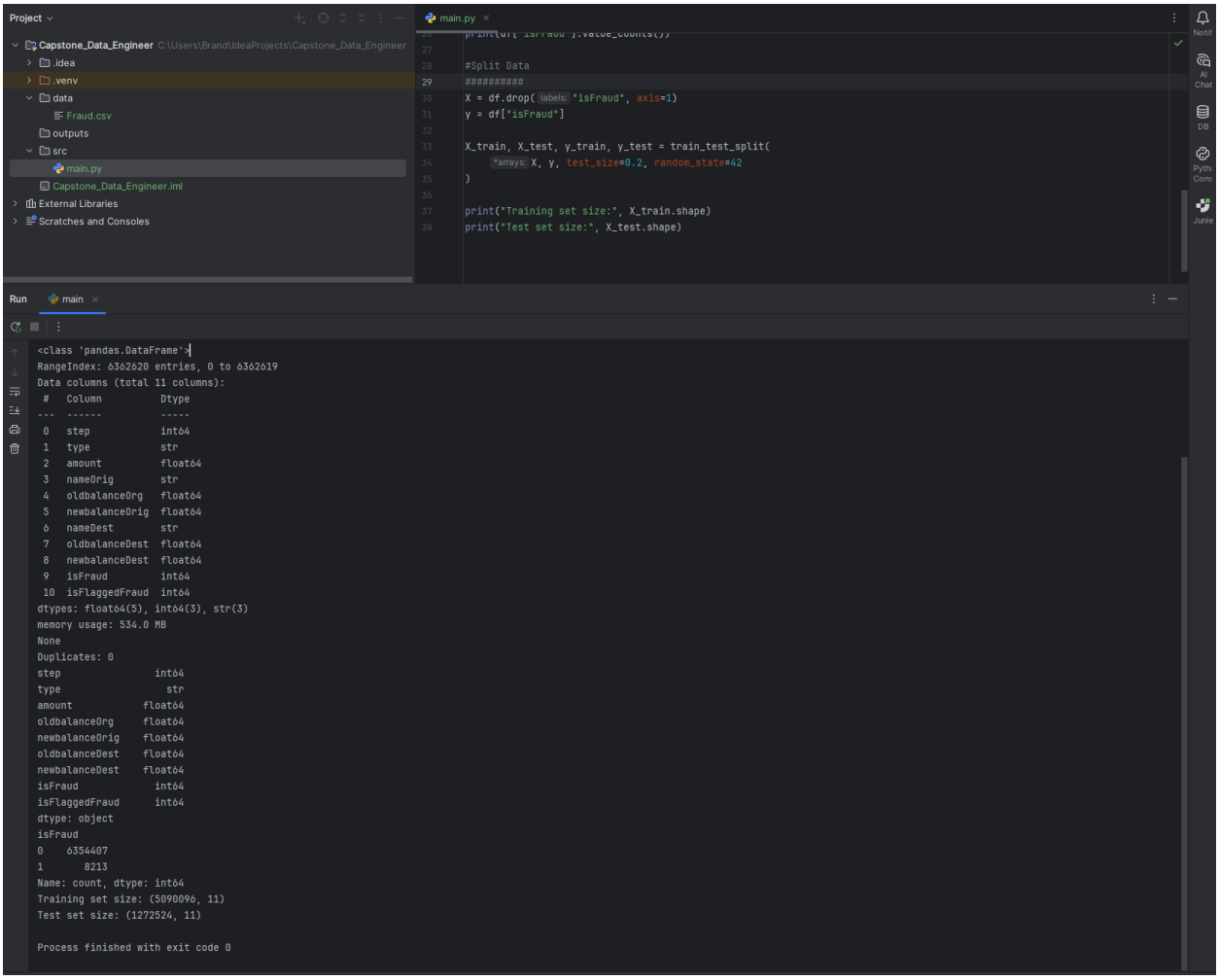


Figure 5. The training and testing dataset split is used for model development and evaluation.

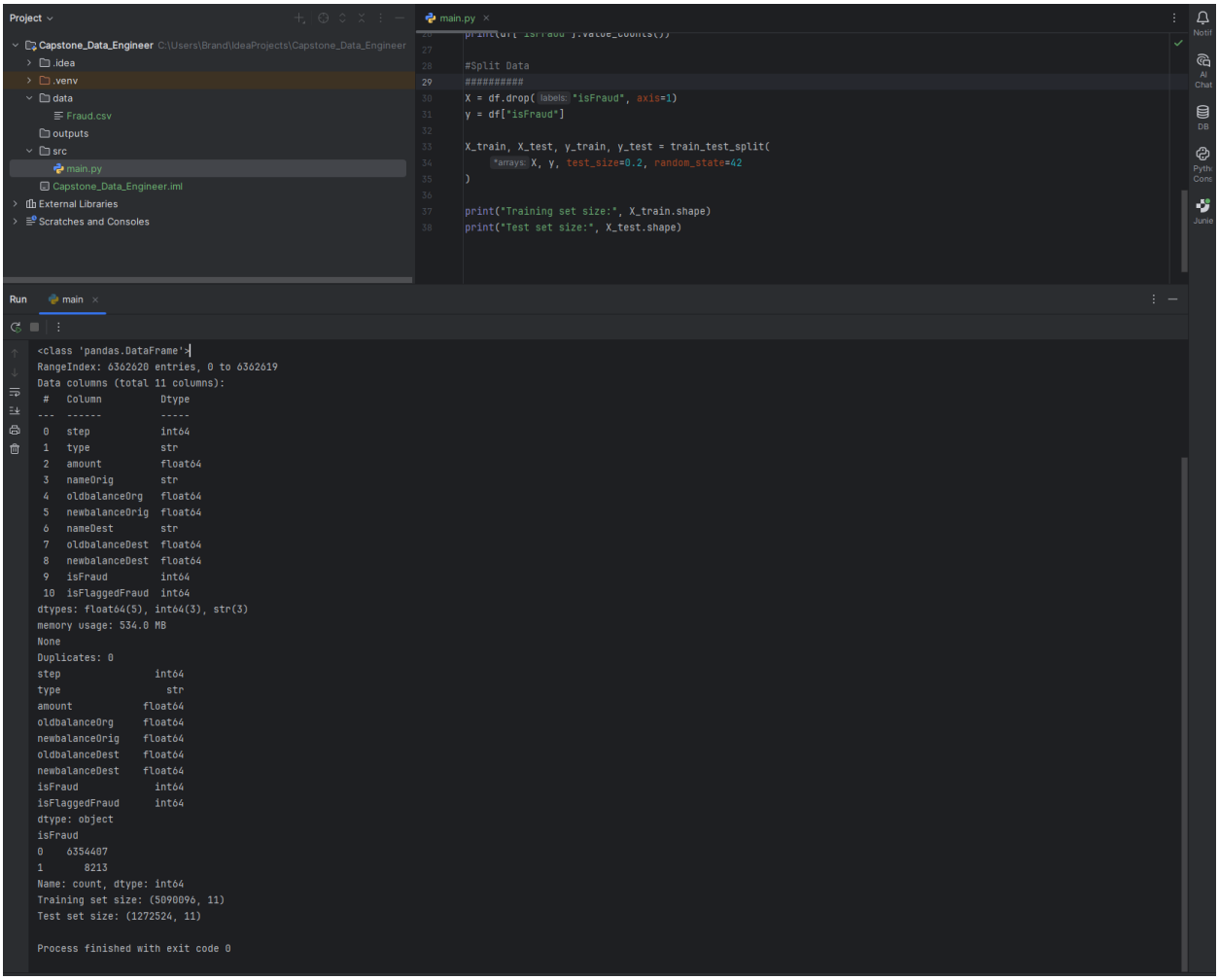


Figure 6. Logistic regression model training completed successfully using the prepared dataset.

The logistic regression model was successfully trained using the training dataset, confirming that the data pipeline and preprocessing steps were correctly implemented.

The model achieved approximately 99.9% accuracy, indicating that most transactions were correctly classified. Precision was approximately 91%, meaning that when the model predicted fraud, it was correct most of the time. However, recall was approximately 48%, indicating that the model identified less than half of the actual fraudulent transactions.

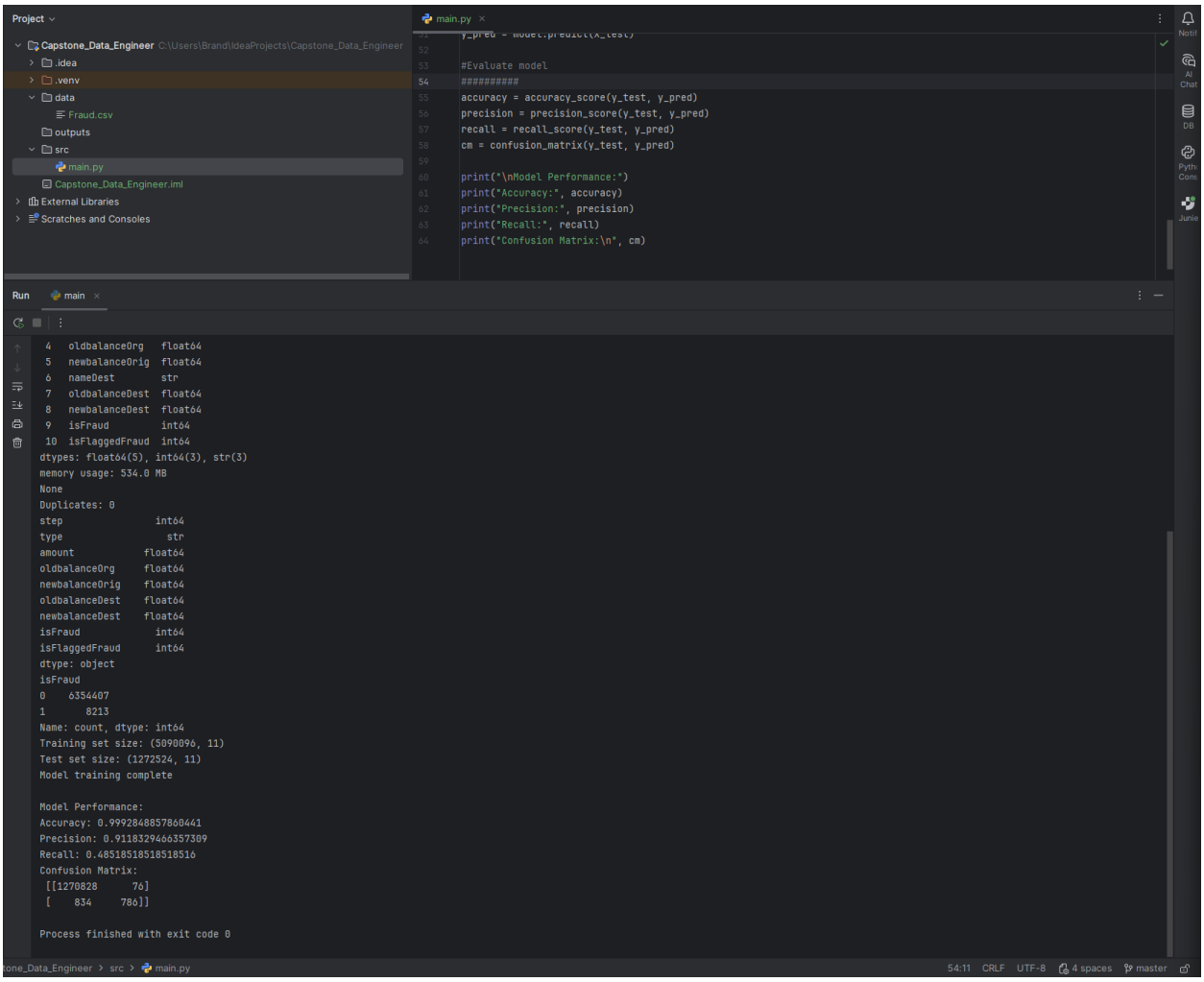


Figure 7. Model performance metrics include accuracy, precision, and recall.

The model performance was evaluated using accuracy, precision, recall, and a confusion matrix to assess its effectiveness in detecting fraudulent transactions.

The confusion matrix showed that a significant number of fraudulent transactions went undetected. In fraud detection, recall is particularly important because failing to detect fraud can result in financial loss.

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Confusion Matrix:
 [[1270828      76]
 [    834     786]]

Process finished with exit code 0
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Figure 8. Confusion matrix showing classification results of the fraud detection model.

The results indicate that while the model performs well at identifying non-fraudulent transactions, it misses a significant number of fraudulent cases, reflected in the relatively low recall score.

The analysis compared model performance to determine which approach provided more effective fraud detection. A decision tree model was also evaluated for comparison. The results indicated that different models may improve fraud detection performance, particularly in terms of recall.

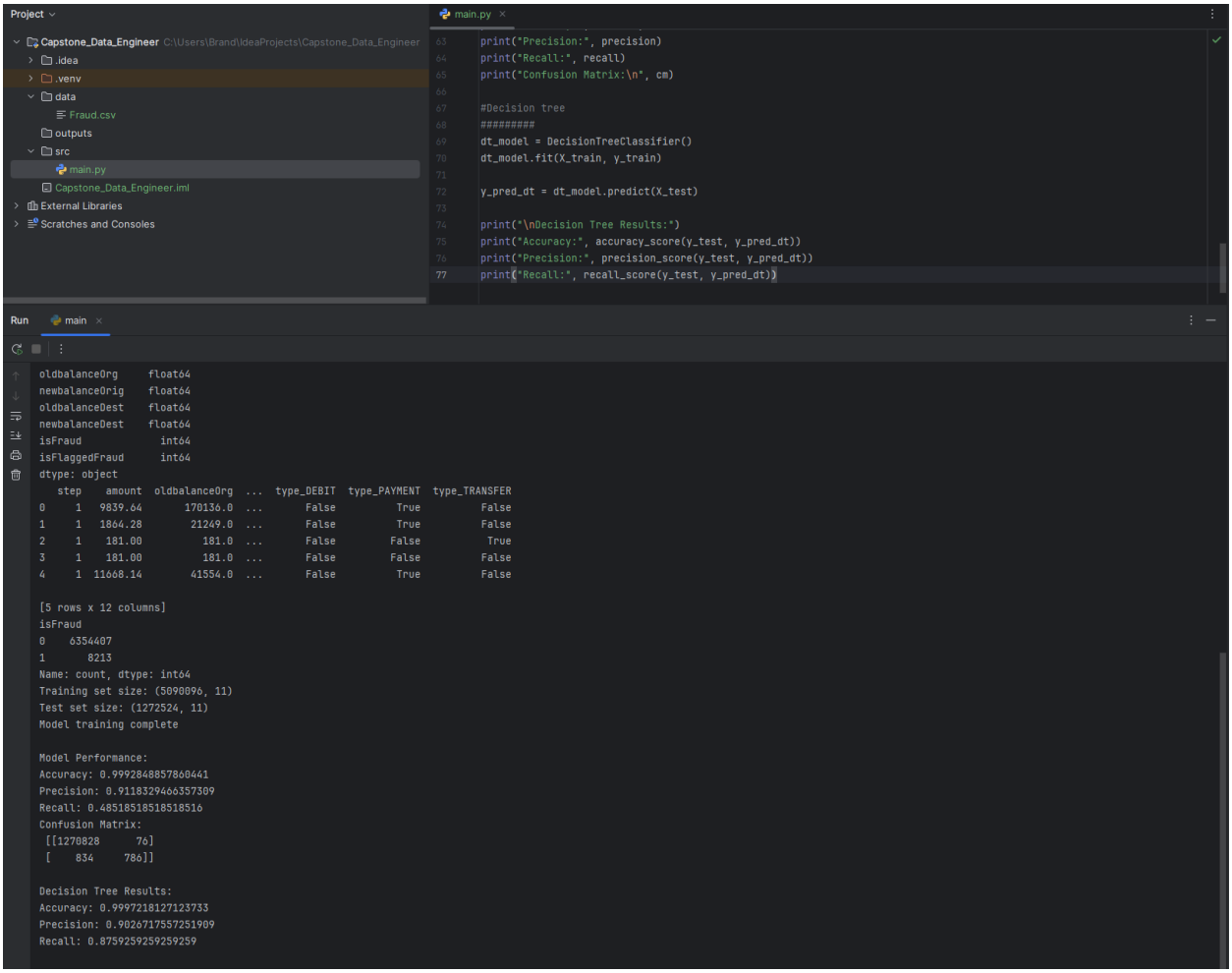


Figure 9. Decision tree model performance metrics showing comparative fraud-detection results with the logistic regression model.

One advantage of logistic regression is its simplicity and interpretability. A disadvantage is that it may not capture complex relationships in the data, limiting its ability to detect all fraudulent activity.

## E. Outcomes and Implications

The analysis results indicate that a scalable data engineering pipeline combined with machine learning can effectively process transactional data and detect fraudulent activity. While the model demonstrated high accuracy and precision, the relatively low recall highlights a limitation in detecting all fraudulent transactions.

In the context of the research question, the findings suggest that data engineering pipelines improve efficiency in processing large datasets. Still, additional improvements in model selection or tuning are necessary to enhance fraud detection performance.

One limitation of this analysis is that the dataset is simulated and may not fully represent real-world financial transaction behavior. Additionally, a simple logistic regression model may limit the ability to capture complex fraud patterns.

Based on these results, it is recommended that financial institutions implement scalable data pipelines combined with more advanced machine learning models to improve fraud detection performance. Improving recall should be prioritized to reduce missed fraudulent transactions.

Future research could explore the use of more complex algorithms, such as random forests or gradient boosting, to improve detection performance. Additionally, future work could focus on implementing real-time data processing systems to detect fraudulent transactions as they occur.

In addition to improving accuracy, the data engineering pipeline increased processing efficiency for large transactional datasets by enabling structured data preparation and faster model training.

## References

Manchanda, C. (n.d.). *Fraudulent transactions data*. Kaggle. https://www.kaggle.com/datasets/chitwanmanchanda/fraudulent-transactions-data

Ngai, E. W. T., Hu, Y., Wong, Y. H., Chen, Y., & Sun, X. (2011). The application of data mining techniques in financial fraud detection. *Decision Support Systems*, 50(3), 559–569.

Chen, M., Mao, S., & Liu, Y. (2014). Big data: A survey. *Mobile Networks and Applications*, 19(2), 171–209.

Kleppmann, M. (2017). *Designing data-intensive applications*. O'Reilly Media.